Mohammad Abdul Mohaimin

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Education

Bangladesh University of Engineering and Technology (BUET) Feb 2020 – Apr 2025
B.Sc. in Mechanical Engineering; *Dhaka, Bangladesh*
Research Interest: Multiphysics modeling, Composite structures, Computational Mechanics, Mechanics of Materials, Numerical Modeling, Structural Integrity, Material Degradation and Damage Mechanics, Failure Analysis, Non-Destructive Testing
GPA: 3.30 out of 4.00 (161 Credits); GPA 4.00 in Undergraduate Thesis (6 Credits)
IELTS Academic*: 7.0 overall (L 7.0, R 8.5, W 7.0, S 6.0) – Oct 2026*

Research Experience

Multiphysics Modeling of Hybrid Laminated Composite Conductors Under a DC Field

Undergraduate Thesis — Supervisor: Dr. Shaikh Reaz Ahmed, Professor, Dept. of Mechanical Engineering, BUET

- Developed a semi-analytical multiphysics framework for hybrid particle-reinforced laminated composite conductors subjected to a direct-current (DC) field, capturing the coupled electro-thermal-mechanical response.
- Formulated the mechanical problem using a Displacement Function approach, reducing the coupled two-dimensional mechanical equilibrium equations to a single fourth-order partial differential equation and enabling a more efficient numerical formulation.
- Implemented a Finite Difference Method (FDM) with iterative solvers to solve the coupled electro-thermal-mechanical system, incorporating nonlinear Joule heating and temperature-dependent electrical resistivity and thermal conductivity.
- Applied the model to Al-SiC and Al-Graphene symmetric laminates with continuous power-law volume-fraction gradation and intermediate bonding layers, and validated voltage, temperature, and stress predictions against COMSOL Multiphysics simulations.

Publications & Presentations

- "Semi-Analytical Multiphysics Framework for Electro-Thermo-Mechanical Response of Functionally-Graded Metal-Matrix Composite Laminates" — M. S. W. Ullah, **M. A. Mohaimin**, T. Islam, M. N. Shahriare, S. R. Ahmed — ***Under Review (Q1 Ranked Journal)***.
- "Assessment of Landfill Gas-to-Energy Potential and Technologies at Arefin Nagar and Fatehabad Landfills Using Site-Specific LandGEM Modeling" — P. Biswangri, A. Chowdhury, A. Mandal, **M. A. Mohaimin** — ***Accepted for presentation (In Dec 2026)***, 8th International Conference on Chemical Engineering (ICCHE), BUET, 2026.

Professional Experience

Bashundhara Oil & Gas Company Ltd. Jan 2026 – Present
Assistant Engineer — Plant Inspection & Integrity

- Identified equipment damage mechanisms and potential failure modes through inspection findings, operating conditions, and historical data, and conducted root-cause analysis to assess material degradation and structural integrity under demanding thermal, mechanical, and chemically aggressive conditions.
- Developed inspection plans based on identified damage and failure mechanisms, and established standard operating procedures (SOPs) for inspection, testing, NDE, and material-quality verification in accordance with API, ASME, and other internationally recognized codes and standards.
- Developing an integrity analysis program for the process equipment by corrosion-rate and remaining-life assessment, translating findings into repair, replacement, and modification recommendations while leading four inspection technicians and coordinating with cross-functional teams.

Abul Khair Group May 2025 – Dec 2025
Technical Management Trainee — Service Management

- Led technical service operations for 7,500+ coffee vending machines and ∼42 service officers across one third of Bangladesh; developed data-driven service-quality reporting and coordinated technical support and spare-parts procurement.

Academic Engineering Projects

Design, Fabrication, and CFD Analysis of a Double-Pipe Heat Exchanger with Disc-Type Turbulator and Outside Fins

- Designed and fabricated a 1-metre counter-flow double-pipe heat exchanger incorporating a 13-disc perforated turbulator and 8 external longitudinal fins for passive heat-transfer enhancement.
- Performed MATLAB-based NTU-effectiveness analysis and 3D CFD analysis in SimScale, demonstrating an additional 2 °C temperature drop due to the turbulator.

Autonomous Detection of Potential Mosquito Breeding Sites Using UAVs and Computer Vision

- Designed, assembled, and field-tested a custom quadcopter UAV integrating an onboard camera, GPS telemetry, flight controller, BLDC motors, and telemetry systems.
- Trained and implemented a YOLOv5 object-detection model on a custom dataset to identify stagnant water and potential mosquito breeding sites from aerial imagery.

Mitigation and Utilization of Methane Emissions from Landfills of Dhaka: Waste-to-Energy Feasibility Study

— Academic Research Project — Supervisor: Dr. Md Aman Uddin, Associate Professor, Dept. of Mechanical Engineering, BUET

- Applied Landfill Gas Emissions Model to estimate methane generation at Matuail and Aminbazar landfills through 2045 using Bangladesh-specific parameters.
- Evaluated electricity-generation potential through anaerobic digestion and incineration and assessed greenhouse-gas mitigation through landfill-gas recovery.

Technical Skills

Programming: MATLAB, Python
Software: COMSOL Multiphysics, ANSYS Workbench, Abaqus, SimScale, SolidWorks, AutoCAD, LaTeX
Engineering: Mechanical Integrity, NDT (UT, PT, MT, VT), Corrosion and Degradation Assessment, Remaining-Life Evaluation, Root Cause Analysis, Welding Inspection, API & ASME codes and standards

Certifications and Courses

- CS50's Introduction to Computer Science - edX - HarvardX
- ISO 9001:2015 QMS & ISO 14001:2015 EMS Internal Auditor (BOGCL, 2026)
- Pressure Vessel Design, Fabrication, Inspection & Testing per ASME BPVC (BUET DCE)